

[DC] Designing Agency-Preserving AI Mediation Supporting Self-Directed Learning in Augmented Reality

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ABSTRACT

AI-mediated assistance is increasingly embedded in learning contexts, yet many systems prioritize efficiency and responsiveness in ways that displace learner control. While prior work on adaptive learning, explainable AI, and intelligent tutoring has examined transparency and personalization, learner agency is typically treated as a secondary design constraint rather than a primary interactional objective. This research investigates how AI-mediated assistance can be designed to preserve learner agency during self-directed learning. It proposes Agency-Preserving AI Mediation (APAM), a design framework that emphasizes negotiable control over when and how assistance is enacted. Using augmented reality (AR) as an interactional medium, the work examines how agency can be made observable through learner action rather than inferred cognitive state. Context-Enhanced Learning (CEL) supports coherent, context-sensitive assistance without asserting learner ability or performance. This work contributes an interaction-centered perspective on agency preservation in immersive AI-mediated learning systems.

Index Terms: agency-preserving AI mediation, self-directed learning, AI-mediated assistance, human-centered AI, augmented reality, context-enhanced learning

1 INTRODUCTION

The increasing availability of chatbots has reshaped how learners seek help during complex tasks, often shifting responsibility for decision-making, strategy selection, and evaluation away from the learner and toward the system. While prior work on intelligent tutoring systems, adaptive learning environments, and explainable AI has focused on improving accuracy, efficiency, or learning outcomes, less attention has been paid to how these systems structure learner control during interaction [1, 4]. As AI assistance becomes more immediate, conversational, and embedded within task contexts, understanding how learner agency is enacted, constrained, or eroded during AI-mediated help-seeking has become an open and consequential research problem. This research seeks to observe how assistance is encountered, negotiated, and enacted when agency is treated as a primary design objective. The following questions foreground learner agency as an interactional phenomenon rather than a cognitive outcome.

RQ1: How is learner agency expressed during interaction with AI assistance when control over the timing, form, and scope of help is made visible and negotiable?

RQ2: What are the boundaries of adaptive user modeling using CEL in supporting coherent, context-sensitive assistance without asserting cognitive ground truth about the learner?

RQ3: How does separating internal schema alignment from external agency negotiation shape learner control during AI-mediated interaction?

RQ4: What aspects of agency signaling, misalignment detection, and assistance negotiation become observable in AR that are difficult to capture in screen-based or conversational AI systems?

2 MOTIVATION AND RELATED WORK

Self-directed learning has long been defined as a process in which learners assume responsibility for goal setting, strategy selection, and self-evaluation [3]. When these responsibilities are externalized to instructional systems, learners may complete tasks efficiently while engaging in limited sensemaking. Recent reviews of AI-supported learning environments indicate that many systems shift regulatory control away from learners by determining when help is provided, what form it takes, and how progress is evaluated [4].

This dynamic is especially pronounced in chatbot-mediated interactions, where assistance is immediate, conversational, and optimized for responsiveness rather than learner agency, reinforcing task completion behaviors and patterns of overreliance [10]. Students often describe these generative AI chatbots as on-demand tutors, despite the absence of pedagogical design or learner control mechanisms [10, 7]. Systems that adapt implicitly based on inferred learner state risk overreliance, misalignment, and reduced learner ownership, even when outcomes appear favorable [5, 6, 8].

Augmented reality (AR) and immersive analytics have been explored as platforms for task guidance and explanation, but prior work has not examined how immersive interfaces might support externally negotiable control over AI-mediated assistance [2, 12]. There is currently no design framework that explicitly treats agency preservation as the organizing principle for AI-mediated assistance, nor work that examines how internal model adaptation can coexist with interaction-level learner control. This research addresses that gap by shifting attention from outcome-based evaluation toward interactional analyses of how learner agency is enacted and preserved during AI-mediated help-seeking.

3 LEARNER AGENCY IN AI-MEDIATED ASSISTANCE

This work conceptualizes learner agency as the capacity to make intentional, consequential choices during learning interactions. This framing is grounded in foundational theories of self-directed learning, which define learning as a process in which individuals retain responsibility for goal setting, strategy selection, and self-evaluation [3]. From this perspective, learning is disrupted when these responsibilities are externalized to AI systems that determine when, how, or whether assistance is enacted.

Agency-Preserving AI Mediation (APAM) is proposed as a design paradigm for AI-mediated assistance that treats agency as a prerequisite for learning rather than a secondary design constraint. In contrast to performance-oriented AI support systems, APAM emphasizes externally negotiable control at the moment assistance is offered. Learners retain authority to accept, reject, modify, or defer assistance in relation to their own goals and understanding,

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rather than being implicitly guided by system-driven adaptation. This framing aligns with research in human–automation interaction demonstrating that effective systems should support appropriate reliance rather than maximal automation [6, 5], as well as human–AI collaboration research showing that even supportive systems shape judgment and reliance through their interactional design [8].

AR is selected as the interactional medium in this work because it uniquely supports the enactment and observation of learner agency during task performance. Unlike screen-based or conversational interfaces that require explicit turn-taking, AR allows AI-mediated assistance to coexist with ongoing activity, enabling learners to encounter assistance peripherally, defer it temporally, or reject it spatially while remaining engaged in the task. Research on situated and embodied cognition emphasizes that sensemaking is shaped by how representations are embedded within action contexts rather than abstracted from them [9, 11].

Within this context, the agency gauge is expected to signal interactions that reflect the balance of control between learner and system during AI-mediated assistance. This design is conceptually descriptive rather than evaluative, with the intent of grounding agency in action rather than inferred cognitive state. The agency gauge is proposed as a hybrid construct that is implicit in sensing but explicit in availability: interaction traces are derived from learner behavior within the AR environment, while the consequences of those traces are made visible and negotiable through interface design. This approach seeks to avoid the automation dominance associated with fully implicit adaptation while preserving continuity of task engagement [8].

APAM Interaction Example in AR

Consider a learner configuring a router in AR who has forgotten the command to set the hostname. The system does not immediately intervene. When the learner pauses or shifts attention toward the configuration panel, a small, spatially anchored indicator appears near the router, signaling that contextual guidance is available without interrupting the task.

The learner may ignore the indicator, move it aside, or approach it to reveal brief, optional guidance grounded in the current configuration step. Suggestions are presented as possible next actions rather than corrective feedback. The learner may adopt, modify, or dismiss them through physical interaction rather than dialog-based confirmation.

Agency is expressed through observable behaviors such as initiating assistance, deferring engagement, selectively applying guidance, or rejecting it altogether. These actions inform the agency gauge without inferring learner ability or intent. Even when system assumptions are imperfect, misalignment becomes visible through continued learner action rather than silent adaptation. The learner retains control of task progression while assistance remains available but non-binding.

4 CONTEXT-ENHANCED LEARNING

Within APAM, Context-Enhanced Learning (CEL) is expected to support internal schema alignment by integrating three distinct sources of contextual information during training and interaction. Each source serves a different representational role and is deliberately bounded to avoid cognitive inference or outcome optimization.

First, a static curriculum captures canonical domain knowledge, including textbook material, formal procedures, and normative task structures. This provides stable conceptual grounding, enabling the model to internalize domain-consistent patterns without tailoring explanations to individual performance or success. Second, a case curriculum represents the current problem context, including task-specific constraints, configurations, and situational variations.

This curriculum allows the system to remain sensitive to the particulars of the learner’s present task without generalizing from outcomes or enforcing predefined solution paths. Third, a user curriculum reflects patterns of observed interaction over time, derived solely from how the learner engages with the system. This includes behaviors such as initiating assistance, deferring engagement, modifying guidance, or rejecting system suggestions. Importantly, this curriculum seeks to encode interactional regularities rather than learner ability, intent, or understanding. CEL integrates domain structure, task context, and interaction history in a manner that aims to preserve SDL without inferring learner cognition or optimizing outcomes [13].

5 EVALUATION AND GOALS

Currently the proposed agency gauge is conceptualized as a hybrid signal derived from observable human-AI interaction patterns rather than cognitive or psychometric inference. Therefore, guidance is sought on validating evaluation strategies that capture agency preservation without collapsing into performance-based metrics. Feedback is also requested on experimental design tradeoffs related to interaction sparsity, misalignment detection, and robustness, as well as on model selection and adaptation strategies for CEL. And, finally, advice is sought on the extent to which agency-preserving mechanisms designed in AR may generalize to other immersive contexts such as virtual reality.

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